DS209: Lecture 8

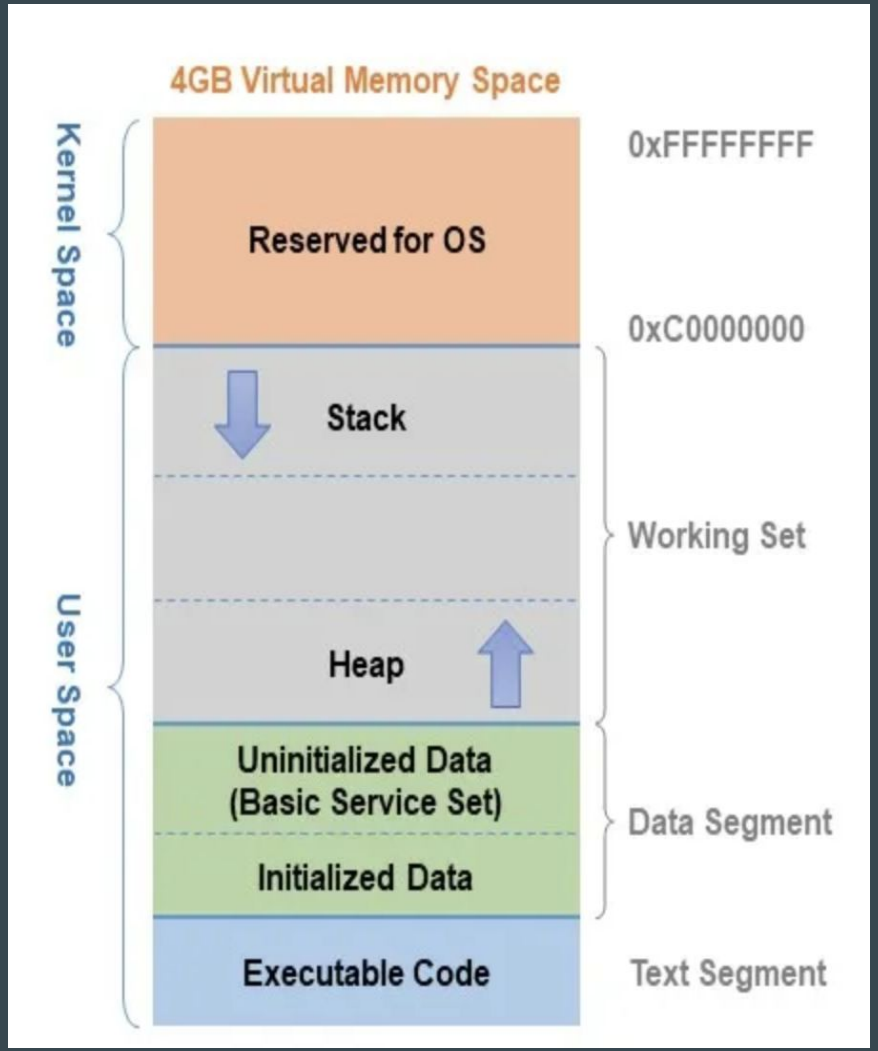


Buffer Overflow

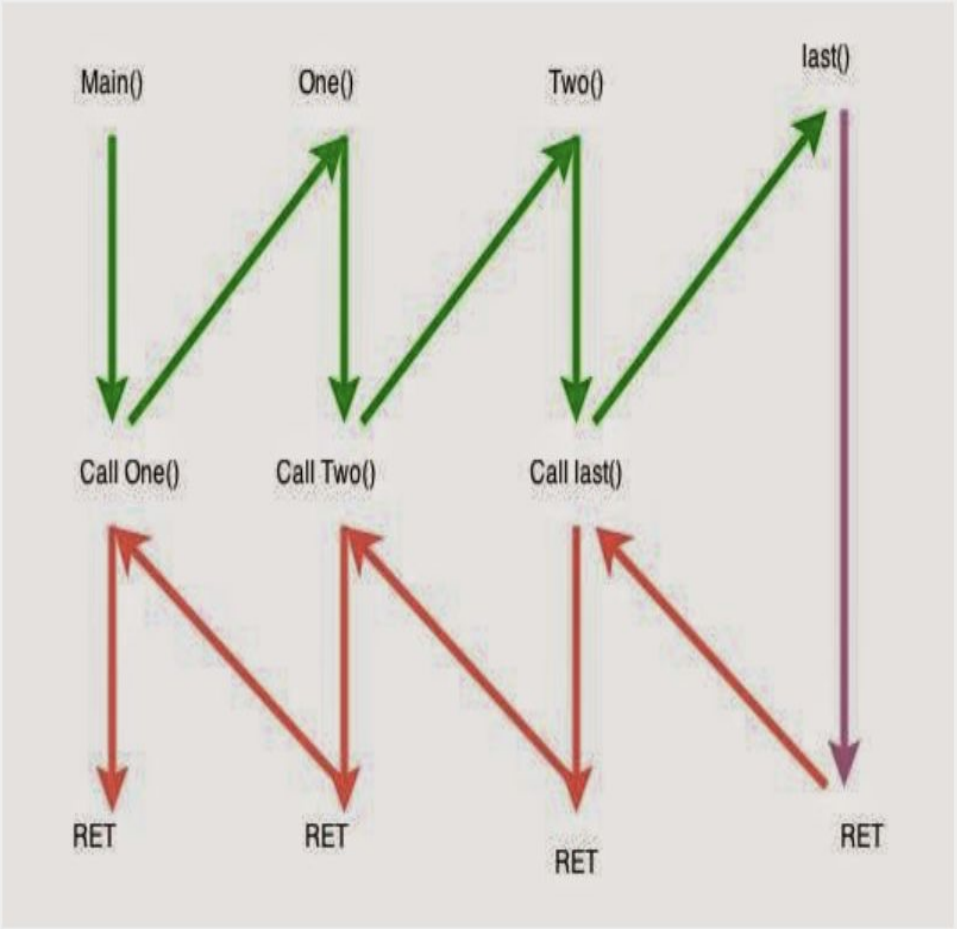
March 7, 2022

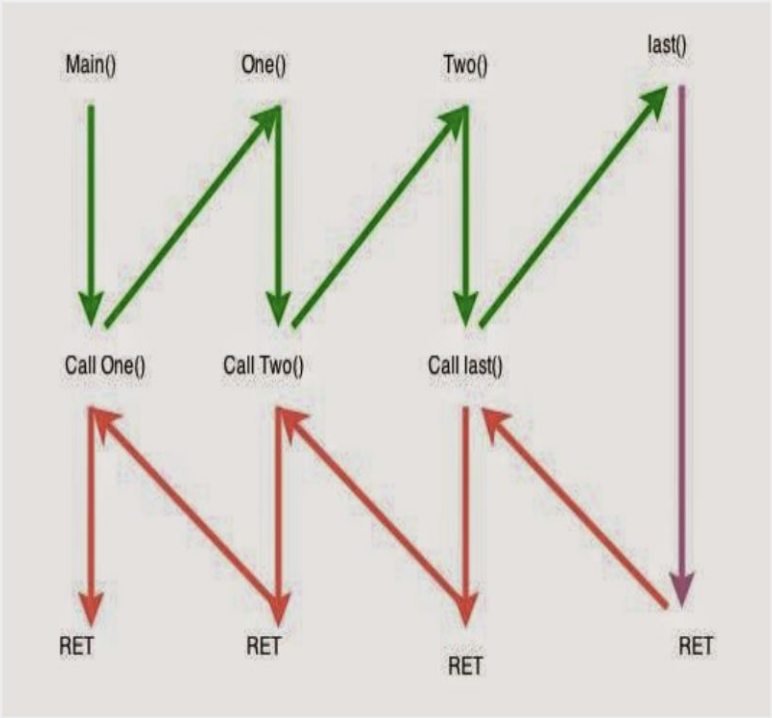
Souradyuti Paul. IIT Bhilai

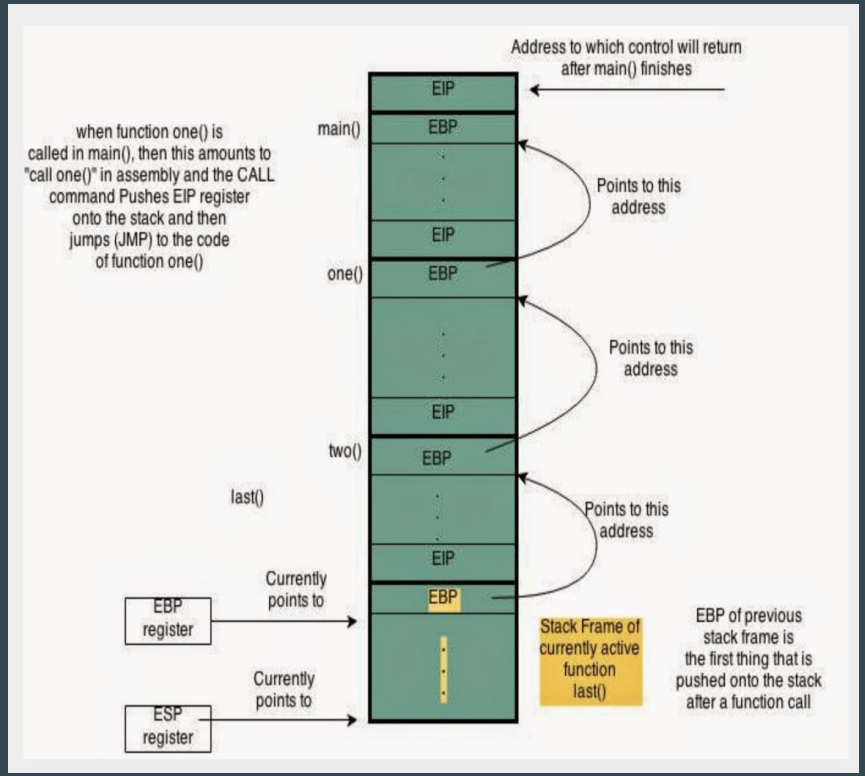
Stack and Heap grow in opposite directions when program runs



Main() calling One(); One() calling Two(); Two calling last()

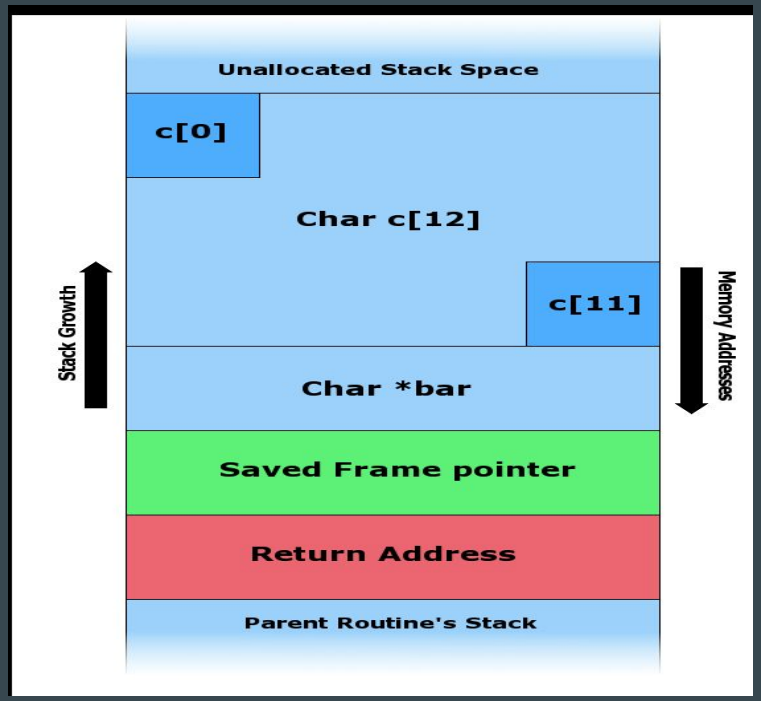






EIP = Return Address

How Stack Grows: Before Data is Copied



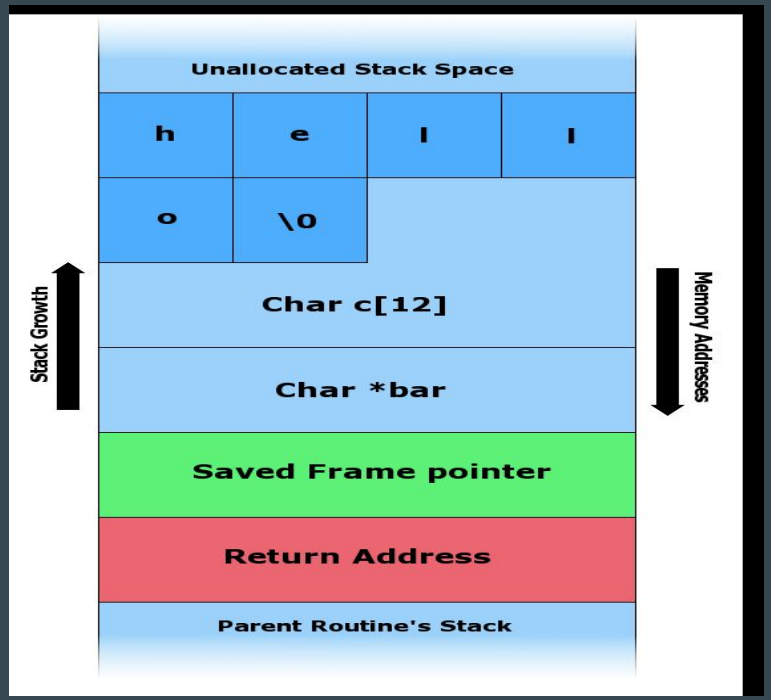

```c
#include <string.h>

void foo(char *bar)
{
   char c[12];

   strcpy(c, bar);  // no bounds checking
}

int main(int argc, char **argv)
{
   foo(argv[1]);
   return 0;
}
```

How Stack Grows: After Data is Copied with argv[1] = “hello”



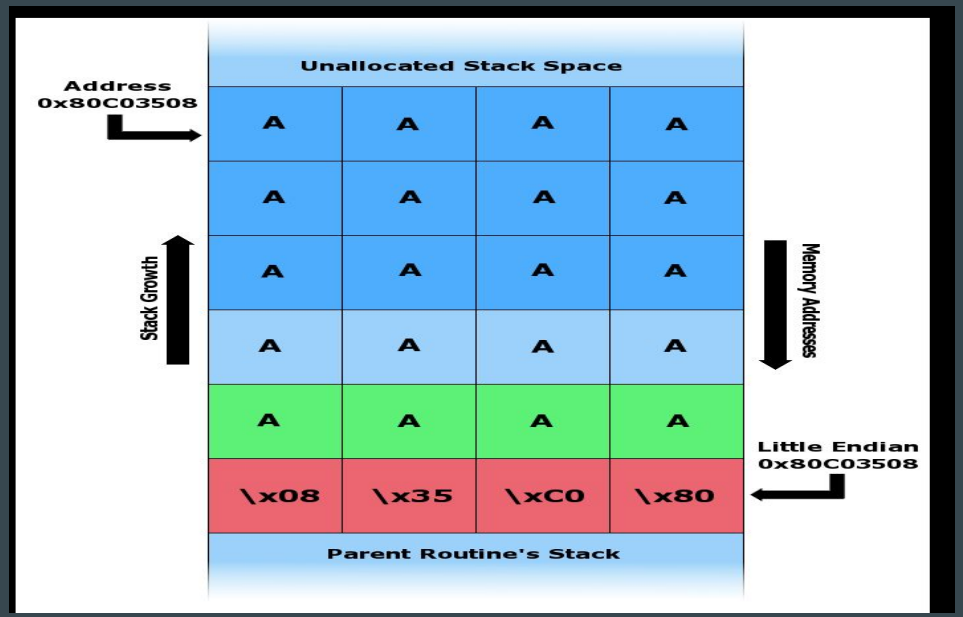

```c
#include <string.h>

void foo(char *bar)
{
   char c[12];

   strcpy(c, bar);  // no bounds checking
}

int main(int argc, char **argv)
{
   foo(argv[1]);
   return 0;
}
```

How Stack Grows: After Data is Copied with argv[1] = “AAA...AAA\x08\x35\xC0\x80” Stack Buffer Overflow Attack!!!




```c
#include <string.h>

void foo(char *bar)
{
   char c[12];

   strcpy(c, bar);  // no bounds checking
}

int main(int argc, char **argv)
{
   foo(argv[1]);
   return 0;
}
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